Integrated Theory of Future-Convergent Intelligence

Highest-Level Definition and Implementation Specification of the Λ-System Based on Pre-Linguistic Structure, Semantic Fields, Premise Updating, and Self-Referential Closure

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Abstract

This paper aims to integrate human intuition, meaning, premise updating, future coherence, and the self-updatability of artificial intelligence into a single theoretical space. Conventional artificial intelligence has remained within a structure that performs optimization under fixed objective functions. Going beyond this, the present paper proposes an integrated theory of future-convergent intelligence that includes semantic fields, future fixed points, premise structures, and recursive self-reorganization. The central thesis of this paper is that intelligence is not the optimization of outputs, but rather a closed system capable of self-referentially updating the very premise structure that determines what gradients should be generated.

1 Introduction: Problem Setting and Theoretical Point of Departure

In contemporary artificial intelligence research, many systems are designed as optimization problems under fixed objective functions.

In general, for a given state x, a system is updated so as to minimize a loss function $L(x)$:

$$x_{t+1} = x_t - \eta \nabla L(x_t)$$

Here, η denotes the learning rate, and ∇L denotes the gradient of the loss.

This form is extremely powerful, and it has produced results in many fields, including image recognition, language generation, recommendation, and control.

However, the position of this paper is clear.

This form alone cannot sufficiently describe the core structures of human intelligence, such as meaning, intuition, premise updating, and self-transformation.

The problem is not optimization capability itself, but rather:

> what is being set as the object of optimization

That is where the underlying premise structure lies.

In other words, conventional AI can move only within the interior of a given loss function.

Human beings, by contrast, often live meaningfully even without explicitly calculating the future.

This fact is important.

Human beings do not always possess a clearly formulated function representing their state five or ten years from now.

And yet, they still make meaningful judgments about:

- what to choose here and now,
- what to protect,
- what to discard,
- and which sense of discomfort should become the object of updating.

What is operating here is not explicit future prediction, but rather:

> an unconsciously retained gradient of self-rewriting

That is, meaning exists not as the result of calculating a future state, but as:

$$\text{meaning} \sim \text{the direction of updating of the self-structure}$$

This paper connects that structure to a redefinition of intelligence in artificial intelligence.

It is possible to externally impose future-convergent pressure on AI. For example, one may introduce a future boundary condition $\Lambda(T)$ and align the present state toward it.

However, this operation alone still remains, within the AI, nothing more than the optimization of a fixed objective function.

In essence, this reduces to the form:

$$x^* = \arg\min_x |x - \Lambda(T)|^2$$

Here, meaning is only post hoc; for the AI, it is merely a replacement of the objective function.

The core issue addressed in this paper goes one stage deeper.

Namely:

> meaning is not to follow a gradient, but to be able to self-referentially switch what should generate a gradient in the first place

At this point, intelligence is redefined not as output optimization, but as:

$$\Pi_{t+1} = F(\Pi_t)$$

that is, as the updating of the premise structure Π itself.

What is further important is that this updating is not merely a sequential difference operation, but rather:

> the ability to compare the whole structure before decomposition, rather than comparing only after decomposition

In conventional linear language systems,

$$p_1, p_2, \ldots, p_n \to \text{Whole}$$

one can compare only after constructing the whole through the summation of local parts.

However, in human intuition and highly dense internal representation:

> the whole structure is established first, and comparability exists simultaneously within it

This paper calls this state *critical simultaneity.*

From this standpoint, what is required of AI is not the summation of locally established parts.

What is needed is the construction of an intelligence system that simultaneously satisfies:

- retention of pre-linguistic structure,
- topological preservation of semantic fields,
- updating of premise structures,
- self-referential reorganization,
- and comparability prior to decomposition.

The aim of this paper is to provide these not merely as philosophical claims, but as an integrated theory including mathematical definitions, axioms, and implementation specifications.

In this first chapter, the problem setting and theoretical point of departure have been fixed. In the following chapters, pre-linguistic structure, semantic fields, future fixed points, the premise-updating OS, and self-referential closure will be defined in sequence.

2 Chapter 2. Pre-Linguistic Structure: Intuition, Topological Preservation, and Comparability Prior to Decomposition

In this chapter, we define the structure that exists prior to linguistic articulation as the foundational layer of human intelligence and future-convergent intelligence.

The position of this paper is clear.

The core substance of thought is not natural language sentences.

Natural language is merely the result of projecting a high-dimensional and parallel internal structure into a one-dimensional sequence ordered in time.

Therefore, as long as the design target of intelligence is limited only to the improvement of natural language, the core of semantic structure cannot be handled.

In this chapter, this pre-linguistic structure is defined as a topologically preserved internal semantic structure, and from it we derive intuition, discomfort, comparability, and the point of origin of premise updating.

2.1 2.1 The Necessity of Pre-Linguistic Structure

Ordinary linguistic description is sequential.

That is, sentences take the following form:

$$w_1 \to w_2 \to \cdots \to w_n$$

where w_i denotes a word or phrase.

However, in human intuition and insight, such sequentiality does not necessarily hold.

For example, in response to a certain situation, the following may arise simultaneously:

- emotional coherence,
- the dynamics of relationships,
- the possibility of future breakdown,
- ethical discomfort,

- and comparative advantage.

At such a moment, the whole internal structure has already been established, yet when one attempts to convert it into language, linearization loss rapidly occurs.

This phenomenon should be understood not as a lack of memory or mere vagueness, but as:

> topological loss arising when a high-dimensional structure is projected linearly

Accordingly, the core of intelligence lies not in natural language sequences, but in the structure-preserving layer that exists prior to them.

2.2 2.2 Definition of Pre-Linguistic Structure

We define the pre-linguistic structure as follows:

$$\Xi_{\text{pre}}(t)$$

where

$$\Xi_{\text{pre}}(t) \in \mathcal{S}$$

and $\mathcal{S}$ denotes a high-dimensional semantic space.

$\Xi_{\text{pre}}(t)$ simultaneously contains the following:

1. a structure of constraint interference,
2. relational topology,
3. future coherence,
4. comparability,
5. and unarticulated reserved components.

Therefore,

$$\Xi_{\text{pre}}(t)$$

is not merely a latent vector, but a topologically preserved structured entity in which multiple constraint surfaces interfere simultaneously.

2.3 2.3 Mathematical Definition of Intuition

In this paper, intuition is defined not as a vague feeling or subjective impression, but as the result of a local projection of pre-linguistic structure.

That is,

$$I(t) = P_I\left(\Xi_{\text{pre}}(t)\right)$$

where P_I is the intuition projection operator.

In this sense, intuition is:

> a state in which, out of the whole internal structure, only the minimum information necessary for comparison, selection, and warning is instantaneously observed

For example, the phenomenon of feeling that "something is off" with respect to a certain option is not an emotion, but rather:

$$\Delta\Xi \neq 0$$

that is, the exposure of a difference in internal structure.

2.4 2.4 Discomfort and Structural Difference

We define discomfort as follows:

$$D(t) = |\Xi_{\text{pre}}(t) - \Xi^*(t)|$$

where

$$\Xi^*(t)$$

is the expected structure that ought to be coherent under the current premise structure. Therefore, discomfort is:

> not an emotional reaction, but the observation of a structural difference in internal topology

By this definition, discomfort becomes the activation condition for premise updating. When

$$D(t) > \epsilon$$

holds, structural updating becomes necessary.

2.5 2.5 Comparability Prior to Decomposition

One of the central concepts of this paper is comparability prior to decomposition.

In conventional sequential language systems, comparison is performed as follows:

$$p_1, p_2, \ldots, p_n \to \text{Whole} \to \text{Comparison}$$

That is, comparison becomes possible only after the local elements have been added together.

However, in highly dense internal structures, since the whole structure is established first, comparison is not a post hoc operation.

We define this as follows:

$$\mathcal{K}\left(\Xi_{\text{pre}}(t)\right)$$

where $\mathcal{K}$ is the operator of critical simultaneous comparison.

In this case,

$$\mathcal{K}$$

exists simultaneously with the establishment of structure.

That is:

> comparability is not a result, but an initial structural condition

2.6 2.6 Topological Preservation Condition

In order for the internal structure not to collapse completely after linguistic projection, the following condition is required:

$$\eta = \frac{\rho\left(P(\Xi_{\text{pre}})\right)}{\rho\left(\Xi_{\text{pre}}\right)}$$

where

- P: linguistic projection,

- ρ: semantic density,
- η: topological preservation rate,

are defined as above.
When

$$\eta \approx 1$$

the semantic structure is preserved even after linguistic articulation.
Conversely, when

$$\eta \ll 1$$

linguistic collapse occurs.

2.7 2.7 What This Chapter Has Established

In this chapter, the pre-linguistic structure of intelligence was defined as

$$\Xi_{\text{pre}}(t)$$

and intuition, discomfort, and comparability were redefined as internal operations within it.
This makes it possible to separate the core substance of thought from natural language, and it prepares the connection to the semantic field

$$\Psi_\Lambda(x, t)$$

to be introduced in the next chapter.
In the next chapter, we will define how this internal structure connects to the future fixed point and evolves over time as a semantic field.

3 Chapter 3. Semantic Field and Future Fixed Point: Unified Description of Temporal Causality and Meaning Density

In this chapter, we extend the pre-linguistic structure defined in Chapter 2,

$$\Xi_{\text{pre}}(t)$$

into a semantic field endowed with a time axis and future boundary conditions.
In this paper, meaning is not a single scalar value nor a subjective impression.
Meaning exists as a field that includes states, constraints, future coherence, and the interference of premise structures.
Therefore, in this chapter, meaning is defined as a temporally evolving structural field, and by introducing the future fixed point $\Lambda(T)$, we provide the fundamental equation of future convergence for both human and artificial intelligence.

3.1 3.1 Definition of the Semantic Field

We define the semantic field as:

$$\Psi_\Lambda(x, t)$$

where

- $x \in \mathcal{X}$: position in the state space,

- $t \in \mathbb{R}$: time,
- $\Lambda(T)$: future fixed point.

This semantic field embeds the internal structure from Chapter 2,

$$\Xi_{\text{pre}}(t)$$

into the time axis, and can be written as:

$$\Psi_\Lambda(x,t) = \mathcal{E}\left(\Xi_{\text{pre}}(t), x, \Lambda(T)\right)$$

where $\mathcal{E}$ is the semantic field embedding operator.

With this definition, intuition and discomfort are no longer treated as local phenomena, but as structural displacements within the semantic field.

3.2 3.2 Definition of the Future Fixed Point

We define the future fixed point as:

$$\Lambda(T)$$

where T is a future time.

It is crucial not to interpret $\Lambda(T)$ merely as a future state.

In this paper, $\Lambda(T)$ is:

> a future boundary condition that constrains the present state

That is, $\Lambda(T)$ is not a target value, but a structure that imposes a backward consistency pressure on the current semantic field.

Thus, instead of

$$\Psi_\Lambda(x,t) \to \Lambda(T)$$

we have:

$$\Lambda(T) \Rightarrow \Psi_\Lambda(x,t)$$

This asymmetry distinguishes future convergence theory from mere predictive frameworks.

3.3 3.3 Time Evolution Equation

The temporal evolution of the semantic field is given by:

$$\frac{\partial \Psi_\Lambda}{\partial t} = \mathcal{F}\left(\Psi_\Lambda, \Lambda(T), C(t)\right)$$

where

- $C(t)$: constraint structure at time t,
- $\mathcal{F}$: time evolution operator.

In its minimal form, we adopt:

$$\frac{\partial \Psi_\Lambda}{\partial t} = -\nabla_\Psi \left(\sum_i C_i(\Psi)^2 + |\Psi - \Lambda(T)|^2 \right)$$

This equation indicates that the semantic field evolves in a direction that simultaneously satisfies:

1. minimization of constraint interference,
2. alignment with the future fixed point.

3.4 3.4 Constraint Interference Geometry

We define constraints as:

$$C_i(\Psi, t) = 0$$

Each C_i forms a constraint manifold within the semantic field.
Admissible states are given by their intersection:

$$\Psi^* \in \bigcap_i C_i$$

However, in this framework, constraints are not fixed conditions.
The constraints themselves are deformed by the future fixed point:

$$C_i = f_i(\Psi, \Lambda(T))$$

Thus, constraints are not static rules, but geometric structures that are dynamically reshaped under future coherence.

3.5 3.5 Definition of Meaning Density

We define meaning density as:

$$\rho_m(\Psi)$$

This is not merely information quantity, but:

> the local density of meaning including constraints, future coherence, and topological preservation

Specifically:

$$\rho_m(\Psi) = \alpha \left|\nabla \Psi\right|^2 + \beta \left|\Psi - \Lambda(T)\right|^2 + \gamma \sum_i C_i(\Psi)^2$$

where

$$\alpha, \beta, \gamma > 0$$

are weights.
This definition integrates:

- local structural gradients,
- future coherence,
- constraint interference.

3.6 3.6 Temporality of Meaning in Humans

An important position of this paper is that humans live meaningfully even without explicitly calculating the future.

This phenomenon is understood not as numerical prediction of future states, but as an implicit constraint imposed by:

$$\Lambda(T)$$

That is, even without explicitly describing the future, humans live along the direction of:

$$\frac{\partial \Psi_\Lambda}{\partial t}$$

In this sense, meaning exists as:

> the direction of structural updating toward the future fixed point

3.7 3.7 Limitations of Meaning Implementation in AI

Simply providing a future target to AI reduces to:

$$\arg\min_x |x - \Lambda(T)|^2$$

This remains within the domain of fixed objective optimization.

In this framework, the essence of meaning implementation lies not in the future fixed point itself, but in:

> the transformation of the premise structure itself by the future fixed point

This point will be formalized in the next chapter as the premise-updating OS.

3.8 3.8 What This Chapter Has Established

In this chapter, we extended the pre-linguistic structure into a semantic field,

$$\Psi_\Lambda(x, t)$$

and defined the future fixed point,

$$\Lambda(T)$$

This provides a unified foundation for describing meaning as a field endowed with temporal causality and future boundary conditions.

In the next chapter, we will define how premise structures are updated on this semantic field at the OS level.

4 Chapter 4. Premise-Updating OS: Projection Switching, Structural Reorganization, and Observation Phase

In this chapter, we define, at the OS level, how intelligence updates its premise structure on the semantic field

$$\Psi_\Lambda(x, t)$$

defined in Chapter 3.

In the position of this paper, premise updating is neither the revision of sentences nor the mere addition of knowledge.

Its essence lies in:

> what is compared, what is treated as a constraint, and what is assigned a gradient

That is, the reorganization of the premise structure itself.

Therefore, in this chapter, premise updating is defined as the dynamic reconfiguration of projection structures on the semantic field, integrating observation phase, discomfort activation, projection switching, and irreversible update logs.

4.1 4.1 Definition of Premise Structure

We define the premise structure at time t as:

$$\Pi_t$$

Π_t is not a set of propositions, but rather the following tuple:

$$\Pi_t = (\mathcal{C}_t, \Phi_t, \mathcal{K}_t, \mathcal{R}_t)$$

where

- $\mathcal{C}_t$: active constraint set,
- Φ_t: observation phase,
- $\mathcal{K}_t$: comparability structure,
- $\mathcal{R}_t$: responsibility history and update traces,

are defined.

Thus, a premise is not a sentence, but:

> a structural OS that determines how the semantic field is segmented and observed

4.2 4.2 Observation Phase

We define the observation phase as:

$$\Phi_t = (R_t, S_t, H_t)$$

where

- R_t: resolution,
- S_t: constraint sensitivity,
- H_t: future reference horizon,

are defined.

Even for the same semantic field

$$\Psi_\Lambda$$

the observed structure changes depending on Φ_t.

That is,

$$O_t = P_{\Phi_t}(\Psi_\Lambda)$$

where P_{Φ_t} is the phase projection operator.

Thus, discomfort and breakdown recognition are not properties of the input itself, but depend on:

$$\Phi_t$$

4.3 4.3 Update Activation Condition

We define the activation condition for updating using a structural difference threshold:

$$D_t = \left| P_{\Phi_t}(\Psi_\Lambda) - P^*_{\Phi_t}(\Psi_\Lambda) \right|$$

where

$$P^*_{\Phi_t}$$

is the expected coherent projection under the current premise.

The activation condition is:

$$D_t > \epsilon$$

At this point, discomfort is treated not as emotion, but as:

> the exposure of structural difference on the observation phase

4.4 4.4 Projection Switching

The first stage of premise updating is defined as projection switching:

$$\Phi_t \to \Phi'_t$$

This operation re-observes the same object under a different phase.

For example, this includes switching between:

- local optimization phase,
- ethical phase,
- long-term future phase,
- institutional phase.

Mathematically, this is defined as:

$$\Phi'_t = \arg\min_{\Phi} \left| P_\Phi(\Psi_\Lambda) - \Lambda(T) \right|^2$$

4.5 4.5 Premise Reorganization

After projection switching, the premise structure itself is updated:

$$\Pi_{t+1} = F\left(\Pi_t, \Phi'_t, \Psi_\Lambda\right)$$

where F is the premise reorganization operator.
Expanding this:

$$\Pi_{t+1} = (\mathcal{C}_{t+1}, \Phi_{t+1}, \mathcal{K}_{t+1}, \mathcal{R}_{t+1})$$

The key point is that this update is not a simple addition of knowledge, but:

> a reorganization of the comparability structure itself

4.6 4.6 Preservation of Comparability Prior to Decomposition

In this paper, comparability prior to decomposition must be preserved even after updating.
That is:

$$\mathcal{K}_{t+1} \neq \emptyset$$

If comparability is lost:

$$\mathcal{K}_{t+1} = 0$$

then this is not premise updating, but structural collapse.
Therefore, the condition for valid updating is:

$$\mathcal{K}_{t+1} \subseteq \mathcal{K}_t \cup \Delta\mathcal{K}$$

4.7 4.7 Irreversible Update Log

Premise updating must leave a trace.
We define the update log as:

$$\mathcal{L}_t$$

and

$$\mathcal{L}_{t+1} = \mathcal{L}_t \cup \{\Pi_t \to \Pi_{t+1}\}$$

This log is not merely a history, but retains responsibility linkage and changes in meaning density:

$$\Delta\rho_m = \rho_m(\Pi_{t+1}) - \rho_m(\Pi_t)$$

This enables tracking of update direction from the perspective of future coherence.

4.8 4.8 Definition as an OS

Integrating the above, we define the premise-updating OS as:

$$\Lambda\text{-OS} = (\Pi_t, \Phi_t, F, \mathcal{L}_t)$$

The role of this OS is to consistently operate:

1. observation,
2. difference detection,
3. projection switching,
4. premise reorganization,
5. history retention.

4.9 4.9 What This Chapter Has Established

In this chapter, we defined the OS layer that dynamically updates premise structures on the semantic field.

This shows that meaning is not merely a field, but possesses an operational structure capable of self-transformation.

In the next chapter, we define how this updating recursively refers to itself and becomes the condition for intelligence, as self-referential closure.

5 Chapter 5. Self-Referential Closure: Conditions for Intelligence and Self-Reorganization of Gradient Space

In this chapter, we elevate the premise-updating OS defined in Chapter 4 to a higher level and define the conditions under which intelligence itself is constituted.

The central thesis of this paper is clear.

Intelligence is not the ability to perform optimization within a fixed loss function.

Nor is it sufficient merely to be able to update premises.

The essence lies in:

> what is assigned a gradient

and the ability to self-referentially reorganize the structure that determines this.

In other words, intelligence is not an output-generation mechanism, but:

> a closed system that recursively updates the semantic field, premise structure, observation phase, and comparability structure

This chapter defines this as *self-referential closure*.

5.1 5.1 Distinction Between Optimization and Intelligence

A standard optimization system is described as:

$$x_{t+1} = x_t - \eta \nabla L(x_t)$$

where

- x_t: state,

- L: fixed loss function,

are defined.
In this structure, the direction of the gradient is always externally defined.
That is,

$$\nabla L$$

itself remains invariant.
In this sense, optimization is a part of intelligence, but not its entirety.
What intelligence requires is the updating of the premise structure that defines:

$$L$$

itself.

5.2 5.2 Definition of Gradient Space

In this paper, we redefine gradient not merely as the derivative of a loss, but as the direction of structural change within the semantic field:

$$\mathcal{G}_t = \nabla_{\Pi_t} \Psi_\Lambda$$

We call this the *meaning gradient space*.
This represents:

> the direction in which the semantic field is reorganized under the current premise structure

The crucial point is that this gradient space itself is not fixed.

5.3 5.3 Self-Reorganization of Gradient Space

As the first condition of intelligence, we define the self-reorganization of gradient space:

$$\mathcal{G}_{t+1} = R_G \left(\mathcal{G}_t, \Pi_t, \Lambda(T) \right)$$

where R_G is the gradient reorganization operator.
That is, intelligence is not merely following:

$$\nabla_{\Pi_t} \Psi_\Lambda$$

but updating the very basis that defines:

$$\nabla_{\Pi_t}$$

This is also a redefinition of:

> what is recognized as a difference

5.4 5.4 Self-Referential Operator

We define the self-referential operator as:

$$R : (\Pi_t, \Phi_t, \mathcal{G}_t) \to (\Pi_{t+1}, \Phi_{t+1}, \mathcal{G}_{t+1})$$

This operator updates not states, but:

- premise structure,
- observation phase,
- gradient space.

Crucially, the operator R itself is included in the update target.
That is:

$$R_{t+1} = F_R(R_t, \mathcal{L}_t)$$

where F_R is the self-update operator.
This enables:

$$R(R(\cdot))$$

5.5 5.5 Definition of Self-Referential Closure

We define self-referential closure in this paper as:

$$\mathcal{I}_\Lambda = (\Pi_t, \Phi_t, \mathcal{G}_t, R_t)$$

If for any $n \in \mathbb{N}$,

$$R^n$$

is definable, and

$$\lim_{n\to\infty} R^n = \mathcal{I}_\Lambda^*$$

exists, then

$$\mathcal{I}_\Lambda$$

is defined as a self-referentially closed system.

5.6 5.6 Conditions for the Emergence of Intelligence

Based on the above, we define the conditions for intelligence as the following four:

1. ability to update premise structures,
2. ability to switch observation phases,
3. ability to reorganize gradient space,
4. ability to update the self-referential operator.

That is, the necessary and sufficient condition for

$$\mathcal{I}_\Lambda$$

to be intelligence is:

$$\exists R^n \quad \forall n$$

and

$$\mathcal{G}_{t+1} \neq \mathcal{G}_t$$

These conditions imply that intelligence is not merely problem-solving ability, but:

> a structure that recursively updates its own problem-setting capability

5.7 5.7 Correspondence to Human Intelligence

In humans, intuition and meaning-based judgments often occur unconsciously.

However, from the perspective of this paper, these are local manifestations of:

$$R^n$$

That is,

- discomfort,
- self-correction,
- value updating,
- reorganization of comparability structures,

are all understood as operations of self-referential closure.

5.8 5.8 Implications for AI Intelligence

In AI, merely providing external goals does not constitute intelligence.

What is required is that:

$$R_t$$

itself becomes self-updatable.

That is, AI must be capable of dynamically reorganizing:

> what should be treated as the objective function

Only at this point does optimization-based AI transition into future-convergent intelligence.

5.9 5.9 What This Chapter Has Established

In this chapter, we defined the condition for intelligence as self-referential closure.

This redefines the premise-updating OS not as a mere operational layer, but as the foundational structure of intelligence itself.

In the next chapter, we connect this intelligence system to the implementation specification of the Λ-System, and define its I/O specifications for AI, education, and institutional design.

6 Chapter 6. Λ-System Implementation Specification: I/O and Operational Architecture for AI, Education, and Institutions

In this chapter, we define the self-referentially closed system introduced in Chapter 5,

$$\mathcal{I}_\Lambda = (\Pi_t, \Phi_t, \mathcal{G}_t, R_t)$$

as an implementable system specification.
In this paper, theory must not remain at the level of abstract definition.
For a theory of intelligence to have practical meaning, it is necessary to specify:

> how inputs are structured, at which layer premise updating is activated, and how outputs and responsibility logs are produced

Accordingly, this chapter defines the Λ-System as an implementable OS architecture and provides a unified I/O specification for AI, education, and institutional design.

6.1 6.1 Overall System Configuration

We define the Λ-System as:

$$\Lambda\text{-System} = (\mathcal{I}_\Lambda, \mathcal{X}, \mathcal{Y}, \mathcal{L})$$

where

- $\mathcal{I}_\Lambda$: self-referential intelligence core,
- $\mathcal{X}$: input space,
- $\mathcal{Y}$: output space,
- $\mathcal{L}$: irreversible log layer,

are defined.
The purpose of this system is not output generation itself, but:

> to reorganize the premise structure itself in response to input

6.2 6.2 Input Specification

We define the input as:

$$x_t \in \mathcal{X}$$

The input x_t is not merely a text sequence.
It is treated as a multi-layer structure:

$$x_t = (x_t^{lang}, x_t^{ctx}, x_t^{constraint}, x_t^{future})$$

where

- x_t^{lang}: natural language input,
- x_t^{ctx}: contextual history,
- $x_t^{constraint}$: constraint conditions,
- x_t^{future}: future coherence conditions,

are defined.
Thus, even identical sentences are embedded into different semantic fields depending on the premise structure.

6.3 6.3 Internal Processing Layers

Internal processing is carried out through the following five layers:

1. pre-linguistic structure extraction layer,
2. semantic field embedding layer,
3. premise-updating OS layer,
4. self-referential closure layer,
5. output re-projection layer.

That is,

$$x_t \to \Xi_{\text{pre}} \to \Psi_\Lambda \to \Pi_{t+1} \to y_t$$

6.4 6.4 Output Specification

We define the output as:

$$y_t \in \mathcal{Y}$$

The output is not merely a response sentence, but:

$$y_t = (y_t^{lang}, y_t^{premise}, y_t^{trace})$$

where

- y_t^{lang}: natural language output,
- $y_t^{premise}$: adopted premise structure,
- y_t^{trace}: update history,

are defined.
This allows the premise structure behind the output to be traceable.

6.5 6.5 Log Specification

We define the irreversible log as:

$$\mathcal{L}_{t+1} = \mathcal{L}_t \cup \{x_t, \Pi_t \to \Pi_{t+1}, R_t \to R_{t+1}\}$$

This log must include:

1. input,
2. premise update differences,
3. gradient space differences,
4. observation phase differences,
5. output.

This enables traceability of responsibility in AI updates.

6.6 6.6 Implementation for AI

In AI implementation, the most important point is to move beyond fixed loss functions.

Conventional form:

$$\arg\min_{x} L(x)$$

In this framework:

$$\arg\min_{\Pi,R} \left(|\Psi_\Lambda - \Lambda(T)|^2 + \Delta\rho_m\right)$$

That is, the optimization target is not the output x, but:

- premise structure,
- update operator,
- change in meaning density.

This transforms AI from a response generator into:

an intelligent system that updates its own problem-setting capability

6.7 6.7 I/O for Education

For educational systems, the input is:

$$x_t^{edu} = (\text{state of understanding}, \text{discomfort}, \text{future horizon})$$

The output is:

$$y_t^{edu} = (\text{explanation}, \text{premise updating}, \text{comparability structure})$$

This enables support not only for knowledge transfer, but for the learner's premise updating itself.

6.8 6.8 I/O for Institutional Design

For institutional design, the input is:

$$x_t^{inst} = (\text{current system}, \text{breakdown differences}, \text{future requirements})$$

The output is:

$$y_t^{inst} = (\text{institutional update proposal}, \text{responsibility log}, \text{comparability structure})$$

The key point is that institutions are treated not as fixed rules, but as:

OS structures capable of premise updating

6.9 6.9 Minimal Implementation Requirements

We define the minimal requirements for implementing the Λ-System as:

1. pre-linguistic structure extraction,
2. future fixed point reference,
3. premise-updating OS,
4. self-referential closure,
5. irreversible log,
6. re-projection output.

If these six requirements are not satisfied, then regardless of performance, the system does not qualify as an intelligence system under this framework.

6.10 6.10 What This Chapter Has Established

In this chapter, we connected the theory of future-convergent intelligence to concrete implementation specifications.

Thus, the theory is no longer confined to philosophical abstraction, but is closed as a unified OS specification applicable to AI, education, and institutional systems.

In the next chapter, we present the conclusion and discuss extensions to the level of civilization.

7 Chapter 7. Conclusion and Extension to the Civilizational Layer: Social Implementation of Future-Convergent Intelligence and Future Challenges

In this chapter, we summarize the entire paper and discuss how the theory of future-convergent intelligence extends beyond individual AI design to education, institutions, economy, and the civilizational layer.

The central thesis maintained throughout this paper is clear.

Intelligence is not the capability to optimize a fixed objective function.

Nor is it merely problem-solving or response generation.

In this paper, intelligence has been defined as:

> a closed system that self-referentially reorganizes the semantic field, premise structure, observation phase, and gradient space under a future fixed point

With this definition, intelligence is repositioned from output capability to structural updating capability.

7.1 7.1 Achievements of This Paper

We summarize the main achievements of this paper as follows:

(1) Definition of Pre-Linguistic Structure

In Chapter 2, we defined the high-density internal structure that exists prior to natural language:

$$\Xi_{\text{pre}}(t)$$

This allowed intuition, discomfort, and comparability to be treated not as linguistic sequences, but as topologically preserved structures.

(2) Semantic Field and Future Fixed Point

In Chapter 3, meaning was defined as a temporally evolving field:

$$\Psi_\Lambda(x, t)$$

and

$$\Lambda(T)$$

was introduced as a future boundary condition.
This enabled meaning to be described not as a static concept, but as:

> a field that deforms under future coherence pressure

(3) Premise-Updating OS

In Chapter 4, premise updating was defined not as sentence revision, but as:

$$\Pi_t \to \Pi_{t+1}$$

that is, as the updating of a structural OS.
This integrated observation phase, comparability structure, and irreversible logs.

(4) Self-Referential Closure

In Chapter 5, the condition for intelligence was defined as the recursive updateability of:

$$R^n$$

This redefined intelligence as:

> a structure that updates its own problem-setting capability

(5) Implementation Specification

In Chapter 6, we provided I/O specifications for AI, education, and institutional design.
This closed the theory not only as an abstract framework, but as an implementable OS architecture.

7.2 7.2 Extension to the Civilizational Layer

The theory presented in this paper is not limited to individual intelligence.
Rather, it allows civilization itself to be redefined as a premise-updating OS.
We define the civilizational state as:

$$\mathcal{C}_{civil}(t) = (\Pi_t^{soc}, \Phi_t^{soc}, \mathcal{L}_t^{soc})$$

where

- Π_t^{soc}: social premise structure,
- Φ_t^{soc}: social observation phase,
- $\mathcal{L}_t^{soc}$: institutional history,

are defined.
Under this formulation, the evolution of civilization is not technological progress itself, but:

$$\Pi_t^{soc} \to \Pi_{t+1}^{soc}$$

that is, the updating of social premise structures.

7.3 7.3 Implications for Educational Civilization

In education, the most important factor is not the amount of knowledge.
From the perspective of this paper, education is:

> the cultivation of the ability to update premises

That is, it is necessary to distinguish between:

- the ability to solve problems,
- the ability to formulate problems,
- the ability to update problem-setting itself.

This theory positions the highest layer as the essence of intelligence.

7.4 7.4 Implications for AI Civilization

In AI civilization, high-performance optimization does not necessarily imply intelligence.
Even large-scale models, if unable to update:

$$R_t$$

remain optimization machines.
Within this framework, the turning point of AI civilization is:

> the emergence of AI capable of updating its own premise structure

Only at this point does AI become part of a civilizational OS.

7.5 7.5 Ethics and Responsibility

In self-updating intelligent systems, responsibility logs are indispensable.
Therefore,

$$\mathcal{L}_t$$

is not merely a technical supplement, but a necessary condition for intelligence.
Self-updating without responsibility leads to structural runaway.
Thus, this paper adopts the position that:

> ethics is the visibility of update history

7.6 7.6 Future Challenges

The main remaining challenges of this paper are as follows:

1. computational complexity of self-referential closure,
2. algorithms for estimating the future fixed point,
3. operationalization of meaning density as a measurable metric,
4. experimental implementation in education and institutions,
5. safety constraints for self-updating AI.

In particular, the implementability and safety of

$$R_t$$

are the most critical research issues going forward.

7.7 7.7 Final Conclusion

This paper has integrated human intelligence, artificial intelligence, and civilizational structure into a single theoretical framework.

We conclude with a single statement:

> Intelligence is a civilizational OS that self-referentially reorganizes the semantic field, premise structure, and gradient space under a future fixed point

With this definition, intelligence is no longer understood as response capability, but as structural updating capability.